

I. Lie Theory

II. Category $\mathcal{O}$

III. KL-Conjecture

$\mathfrak{g}$: semisimple Lie algebra, f.d. $\mathbb{C}$. $\mathfrak{h}$: Cartan subalg.

$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$, where $\mathfrak{g}_\alpha := \{x \in \mathfrak{g} \mid [h, x] = \lambda(h)x \ \forall h \in \mathfrak{h}\}$.

$\mathfrak{h}^* \supset \Phi := \{\alpha \in \mathfrak{h}^* \mid \alpha \neq 0, \mathfrak{g}_\alpha \neq 0\} \supset \Phi^+ \supset \Pi$, $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$, $b = \mathfrak{h} \oplus \mathfrak{n}_+$

The Killing form $K(x, y) = \text{tr}(\text{ad } x \circ \text{ad } y)$ is nondegenerate on $\mathfrak{h}$

$\leadsto$ inner product $(\cdot, \cdot)$ on $\text{span}_{\mathbb{R}} \Phi = E \subset \mathfrak{h}^*$.

Define the reflection $S_\alpha: E \rightarrow E$ by $S_\alpha(\beta) := \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \alpha$.

The Weyl group $W := \langle S_\alpha \mid \alpha \in \Pi \rangle$. $S = \{S_\alpha \mid \alpha \in \Pi\}$,

(W, S) is a finite crystallographic Coxeter system. i.e. $\forall s, t \in S$, $m_{s,t} \in \{2, 3, 4, 6\}$.

$\mathfrak{g}\text{-mod} \longleftrightarrow \mathcal{U}(\mathfrak{g})\text{-mod}$, $\mathcal{U}(\mathfrak{g}) = T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] \rangle$.

II. Def. The BGG category $\mathcal{O}$ is the full subcat. of $\mathcal{U}(\mathfrak{g})\text{-mod}$ whose objects are modules satisfying:

(O1). M is a f.g. $\mathcal{U}(\mathfrak{g})\text{-mod}$.

(O2). $M = \bigoplus_{\lambda \in \mathfrak{h}^*} M_\lambda$, where $M_\lambda := \{v \in M \mid h \cdot v = \lambda(h)v \ \forall h \in \mathfrak{h}\}$.

(O3). $\forall v \in M$, $\mathcal{U}(\mathfrak{n}_-)\cdot v$ is f.d. $\text{wt}(M) := \{\lambda \in \mathfrak{h}^* \mid M_\lambda \neq 0\}$.

Fact: $\{\text{f.d. } \mathcal{U}(\mathfrak{g})\text{-mod}\} \subset \mathcal{O}$.

Def. Let $M \in \mathcal{O}$. A nonzero vector $v^+ \in M$ is called a maximal vector of wt. $\lambda \in \mathfrak{h}^*$ if $v^+ \in M_\lambda$ and $\mathfrak{n}_+ \cdot v^+ = 0$.

Def. A module $M \in \mathcal{O}$ is called a highest weight module of wt. λ if $M = \mathcal{U}(\mathfrak{g}) \cdot v^+$ for some max. vector $v^+ \in M_\lambda$.

Prop. M : h.t. wt. mod. of wt. λ .

(a) $\text{wt}(M) \subseteq \lambda - \mathbb{N}\Phi^+$.

(b) $\forall \mu \in \mathfrak{h}^*$, $\dim M_\mu < \infty$. $\dim M_\lambda = 1$.

(c) M has a unique max. submod.

unique simple quotient.

(d) All simple ht. wt. mod. of wt. λ are iso.
 $\mathfrak{h}^* \longleftrightarrow \{\text{simple ht. wt. mod.}\}$

Thm. $\forall M \in \mathcal{O}$, M has a finite filtration s.t. each quotient is a ht. wt. mod.

pf. Let S be a finite generating set of M , consisting of weight vectors. By Lie's thm, as $U(\mathfrak{b})$ -mod., $\exists 0 = V_0 \subset V_1 \subset \dots \subset V_n = U(\mathfrak{b}) \cdot S$ s.t. V_i/V_{i-1} is 1-dim. Then each quotient of $0 = U(\mathfrak{n}^-)V_0 \subset U(\mathfrak{n}^-)V_1 \subset \dots \subset U(\mathfrak{n}^-)V_n = U(\mathfrak{g}) \cdot S = M$ is a ht. wt. m.

Def. $\forall \lambda \in \mathfrak{h}^*$, define an 1-dim $U(\mathfrak{b})$ -mod. $\mathbb{C}_\lambda = \mathbb{C} \cdot v_\lambda$ by $h \cdot v_\lambda = \lambda(h) v_\lambda, \forall h \in \mathfrak{h}, \mathfrak{n} \cdot v_\lambda = 0$.

Set $\Delta(\lambda) := U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$, called the Verma module.

$\Delta(\lambda) \cong U(\mathfrak{g})/I$, $I = \langle \mathfrak{n}, h - \lambda(h) \cdot 1, h \in \mathfrak{h} \rangle$.

If $M = U(\mathfrak{g}) \cdot v^+$, with $v^+ \in M_\lambda$, then $I \cdot v^+ = 0$. Hence $\Delta(\lambda) \twoheadrightarrow M$.
Denote $L(\lambda)$ the unique simple quotient of $\Delta(\lambda)$.

$M = U(\mathfrak{g}) \cdot v^+, v^+ \in M_\lambda, z \in Z(\mathfrak{g})$, center of $U(\mathfrak{g})$

$h \cdot (z \cdot v^+) = z \cdot (h \cdot v^+) = \lambda(h) z \cdot v^+ \Rightarrow z \cdot v^+ \in M_\lambda$

$\Rightarrow z \cdot v^+ = \chi_\lambda(z) v^+$ for some $\chi_\lambda(z) \in \mathbb{C}$.

This defines an alg. hom. $\chi_\lambda: Z(\mathfrak{g}) \rightarrow \mathbb{C}$, called the central character.

Thm (Harish-Chandra): $\chi_\lambda = \chi_\mu \Leftrightarrow \mu = w \cdot \lambda, w \in W$
 $= w(\lambda + \rho) - \rho$, where
 $\rho = \frac{1}{2} \sum_{\alpha \in \check{\Sigma}^+} \alpha$.

For $M \in \mathcal{O}$, define $M^{\chi_\lambda} := \left\{ v \in M \mid (z - \chi_\lambda(z))^{n_z} \cdot v = 0, n_z \in \mathbb{N}, \forall z \in Z(\mathfrak{g}) \right\}$

Define $\mathcal{O}_\lambda = \mathcal{O}_{\chi_\lambda}$ to be the full subcat. of $\mathcal{O}$ consisting of $M \in \mathcal{O}$ s.t. $M = M^{\chi_\lambda}$.

$$\Delta(w \cdot \lambda) \in \mathcal{O}_\lambda. \quad \forall w \in W.$$

We have the decomp. $\mathcal{O} = \bigoplus_{\lambda \in \mathfrak{h}^* / (W, \cdot)} \mathcal{O}_\lambda$.

If λ is integral, i.e., $(\lambda, \alpha^\vee) \in \mathbb{Z} \quad \forall \alpha \in \Phi$, then $\mathcal{O}_\lambda$ is indecomp.

Def. We call $\mathcal{O}_0 = \mathcal{O}_{\chi_0}$ the principal block.

Thm. $\mathcal{O}$ is Artinian.

pf. $M \in \mathcal{O} = \bigoplus_{\lambda \in \mathfrak{h}^* / (W, \cdot)} \mathcal{O}_\lambda$. Assume $M = \Delta(\lambda)$.

$$M = M_0 \supset M_1 \supset \dots, \quad M_i / M_{i+1} \cong L(w \cdot \lambda), \quad w \in W.$$

$$\dim M_{w \cdot \lambda} < \infty \Rightarrow \text{DCC}.$$

If λ is dominant, i.e., $(\lambda, \alpha^\vee) \in \mathbb{Z}_{\geq 0}$, then for $M \in \mathcal{O}_\lambda$, $\text{Hom}_{\mathcal{O}}(\Delta(\lambda), M) \cong \text{Hom}_{\mathcal{U}(\mathfrak{b})}(\mathbb{C}\lambda, M) \cong M_\lambda$.

$\Rightarrow \text{Hom}_{\mathcal{O}}(\Delta(\lambda), \cdot)$ exact $\Rightarrow \Delta(\lambda)$ is proj.

If $E \in \mathcal{O}$ is f.d., then $\text{Hom}_{\mathcal{O}}(\Delta(\lambda) \otimes E, \cdot) \cong \text{Hom}_{\mathcal{O}}(\Delta(\lambda), E^* \otimes \cdot)$ is exact. $\Rightarrow \Delta(\lambda) \otimes E$ is proj.

For every $\lambda \in \mathfrak{h}^*$, we can choose a dominant wt. μ & a f.d. $\mathcal{U}(\mathfrak{g})$ -mod. E s.t. $\text{Hom}_{\mathcal{O}}(\Delta(\mu) \otimes E, L(\lambda)) \neq 0$.

$\Rightarrow \mathcal{O}$ has enough proj. (Since $L(\lambda)$'s are all the simple objects in $\mathcal{O}$).

Every $M \in \mathcal{O}$ has a proj. cover $\pi: P_M \rightarrow M$, where π is surj. essential, i.e. no proper submod. of P_M is mapped onto M .

When $M = L(\lambda)$, we write $P_M = P(\lambda)$.

$\pi_\lambda: P(\lambda) \twoheadrightarrow L(\lambda)$ is essential $\Rightarrow \text{Ker } \pi$ is the unique max. submod. of $P(\lambda) \Rightarrow P(\lambda)$ is indecomp.

$$\Delta(\mu) \otimes E \twoheadrightarrow L(\lambda). \Rightarrow \Delta(\mu) \otimes E = P(\lambda) \oplus \sim$$

Def. $M \in \mathcal{O}$. Recall that M has a finite filtration with quotients being ht. wt. mod. If each quotient is a Verma mod., then we call the filtration a Verma flag (or a std. filtration).

Denote the flag multiplicity by $(\mu: \Delta(\lambda))$.

$$\Delta(\mu) \otimes E = (\mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{b})} \mathbb{C}_\mu) \otimes_E E \cong \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{b})} (\mathbb{C}_\mu \otimes E).$$

$\Rightarrow \Delta(\mu) \otimes E$ has a Verma flag.

$\Rightarrow P(\lambda)$ has a Verma flag

$\Rightarrow$ every proj. has a Verma flag. ($P(\lambda)$ are all indecomp. proj. in $\mathcal{O}_\lambda$)

BGG reciprocity: $(P(\lambda): \Delta(\mu)) = [\Delta(\mu): L(\lambda)]$.

II Define $\text{ch } M = \sum_{\mu \in \mathfrak{h}^*} (\dim M_\mu) e^\mu$, called the (formal) character

e.g. $\text{ch } \Delta(\lambda) = \frac{e^\lambda}{\prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})}$.

Question: compute $\text{ch } L(\lambda)$.

When λ is dominant integral, $L(\lambda)$ is f.d., and we have the Weyl character formula: $\text{ch } L(\lambda) = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{(w \cdot \lambda)}}{\prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})}$

Thm (Verma). $\lambda \in \mathfrak{h}^*$, dominant. Suppose $x, y \in W$ s.t. $x \geq y$ in the Bruhat order. Then $\text{Hom}_{\mathcal{O}}(\Delta(x \cdot \lambda), \Delta(y \cdot \lambda)) \cong \mathbb{C}$.
 $\Rightarrow [\Delta(y \cdot \lambda): L(x \cdot \lambda)] \neq 0$.

In general, $\dim \text{Hom}_{\mathcal{O}}(\Delta(\lambda), \Delta(\mu)) \leq 1$.

Thm (BGG) If $\text{Hom}_{\mathcal{O}}(\Delta(\mu), \Delta(\eta)) = 1$, then $\exists \lambda$ dominant s.t. $\mu = x \cdot \lambda$, $\eta = y \cdot \lambda$ with $x \geq y$ in the Bruhat order.

Note: $[\Delta(y \cdot \lambda): L(x \cdot \lambda)] \neq 0 \Rightarrow x \geq y$.

Since $[\Delta(y \cdot \lambda): L(y \cdot \lambda)] = 1$, we have $\{[\Delta(y \cdot \lambda)] \mid y \in W\}$ and $\{[L(y \cdot \lambda)] \mid y \in W\}$ are both bases for the Grothendieck group for $\mathcal{O}_\lambda$.

Kazhdan-Lusztig Conjecture: λ : regular, integral, dominant.

$$[\Delta(y \cdot \lambda)] = \sum_{x \geq y} h_{y,x}(1) [L(x \cdot \lambda)], \text{ where } h_{y,x} \text{ is given by}$$
$$b_x = \sum_{y \geq x} h_{y,x}(v) \delta_y, \text{ with } b_x: \text{KL-basis, } \delta_x: \text{std basis for } H(W)$$